Grade 10 – Term 1
Decision-Making Under Uncertainty
Exam Solutions

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Learning objective

Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.

Criteria assessment

This exam assesses C8 and C9. Answer individually on separate paper. Questions 1–2 assess C8; each question is worth 3 points: 1 point for task (a) and 2 points for task (b). Questions 3–5 assess C9; each problem is worth 2 points: 1 point for the filtering tree and interpretation and 1 point for the supporting table. Each criterion totals 6 points. A **minimum of 4 points** is required to pass each criterion. Partial credit is available for correct comparisons, filtering stages, and justified reasoning even if later conclusions contain errors.

Evaluated criteria

C8 Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations.

C9 Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

1 Repair Workshop Plans [C8-3]

A bicycle repair workshop chooses next quarter's capacity before demand is known. Profits are in thousand USD; forecast probabilities are shown below.

Alternative	High	Normal	Low
$P(S_j)$	0.2	0.6	0.2
Large (A_1)	120	25	-20
Flexible (A_2)	85	65	10
Basic (A_3)	50	45	35
Partner (A_4)	70	75	20

Decision statistics (Regret means maximum regret):

Alternative	Max	Min	Regret	EMV
A_1	120	-20	55	35
A_2	85	10	35	58
A_3	50	35	70	44
A_4	75	20	50	63

Questions/Tasks

C8 Report each criterion's recommended alternative(s), deciding statistic, and value. **[1]**

C8 A subsidy s (integer, $0 \le s \le 30$, thousand USD) increases the low-demand payoffs of both A_1 and A_2 by s. All other payoffs and probabilities stay fixed. An alternative is eligible only if its profit is nonnegative in every state. Determine the ranges of s for which each alternative is eligible. Find the smallest s that changes the Maximin recommendation from part (a) to A_2 alone. At that s, report all four recommendations and deciding values among eligible alternatives. **[2]**

(a) Original recommendations [1]

Criterion	Recommended alternative(s)	Deciding statistic and value
Maximax	A_1	Row maximum: 120
Maximin	A_3	Row minimum: 35
Minimax Regret	A_2	Maximum regret: 35
EMV/Bayes	A_4	EMV: 63

All deciding values are in thousand USD. Maximax pursues the greatest upside; Maximin protects the worst outcome; Minimax Regret limits missed profit; EMV uses the forecast probabilities to maximize expected profit. Regret benchmarks use all four original alternatives, with state bests $(120, 75, 35)$; no eligibility restriction applies in (a).

(b) Subsidy and eligibility **[2]**

The changed low-demand payoffs are $-20+s$ and $10+s$. Over the entire allowed integer range:

Alternative	Nonnegative-profit condition	Eligible integer subsidies
A_1	$-20+s \geq 0$	$20 \leq s \leq 30$
A_2	$10+s \geq 0$	$0 \leq s \leq 30$
A_3	Minimum $35 \geq 0$	$0 \leq s \leq 30$
A_4	Minimum $20 \geq 0$	$0 \leq s \leq 30$

Thus the eligible set is $\{A_2, A_3, A_4\}$ for $0 \leq s \leq 19$ and all four alternatives for $20 \leq s \leq 30$. The row minima over this range are

$$m_1(s) = -20+s, \quad m_2(s) = 10+s, \quad m_3 = 35, \quad m_4 = 20.$$

These formulas hold because the changed low payoff remains the smallest in each affected row. To select A_2 alone requires $10+s > 35$, hence $s > 25$. At $s = 25$, A_2 and A_3 tie at 35; for $s \leq 24$, A_3 wins. Since $m_1 \leq 10$ and $m_4 = 20$, neither can obstruct the strict win. Therefore the smallest integer subsidy is $s = 26$, or \$26,000.

At $s = 26$, all four alternatives are eligible. The updated payoffs and statistics (thousand USD) are:

	High	Normal	Low	Max	Min	Max regret	EMV
A_1	120	25	6	120	6	50	40.2
A_2	85	65	36	85	36	35	63.2
A_3	50	45	35	50	35	70	44
A_4	70	75	20	75	20	50	63

Regret benchmarks are computed over the eligible set $\{A_1, A_2, A_3, A_4\}$ using the changed payoffs: $b = (120, 75, 36)$. Subtract each payoff from its state benchmark:

$$r_1 = (0, 50, 30), \quad r_2 = (35, 10, 0), \quad r_3 = (70, 30, 1), \quad r_4 = (50, 0, 16).$$

Their largest entries give the maximum regrets above. The affected EMVs are

$$EMV(A_1) = 0.2(120) + 0.6(25) + 0.2(6) = 40.2,$$

$$EMV(A_2) = 0.2(85) + 0.6(65) + 0.2(36) = 63.2.$$

The other EMVs remain 44 and 63. Consequently:

Criterion	Recommended alternative(s)	Deciding statistic and value
Maximax	A_1	Row maximum: 120
Maximin	A_2	Row minimum: 36
Minimax Regret	A_2	Maximum regret: 35
EMV/Bayes	A_2	EMV: 63.2

2 Community Market Stall [C8-3]

An organizer chooses a weekend market stall. Profits are in hundred USD; forecast probabilities are shown below.

Alternative	High	Low
$P(S_j)$	0.4	0.6
Large (C_1)	80	10
Compact (C_2)	48	32

Alternative	Max	Min	Regret	EMV
C_1	80	10	22	38
C_2	48	32	32	38.4

Questions/Tasks

C8 Report each criterion's recommended alternative(s), deciding statistic, and value. **[1]**

C8 A sponsor guarantees C_2 a high-attendance profit of g, replacing 48, where g is an integer, $48 \leq g \leq 80$, in hundred USD. All other payoffs and probabilities stay fixed; both stalls remain eligible. Find the smallest g that changes the Minimax Regret recommendation from part (a) to C_2 alone. Justify the threshold and boundary tie; interpret the increase from 48 in USD. At that g, report all four updated recommendations and deciding values, including every tie. **[2]**

(a) Original recommendations [1]

Criterion	Recommended alternative(s)	Deciding statistic and value
Maximax	C_1	Row maximum: 80
Maximin	C_2	Row minimum: 32
Minimax Regret	C_1	Maximum regret: 22
EMV/Bayes	C_2	EMV: 38.4

All values are in hundred USD. Regret benchmarks use both original stalls, with state bests $(80, 32)$. Only EMV uses the probabilities; the other criteria compare upside, downside, or missed profit.

(b) Guaranteed profit threshold **[2]**

For every allowed integer $48 \le g \le 80$, both stalls are eligible. Using the full eligible set $\{C_1, C_2\}$, the state bests remain $(80, 32)$. Thus

$$r(C_1) = (0, 22), \quad R(C_1) = 22, \qquad r(C_2) = (80 - g, 0), \quad R(C_2) = 80 - g.$$

For C_2 alone to minimize maximum regret,

$$80 - g < 22 \iff g > 58.$$

At $g = 58$, both maximum regrets equal 22, so both stalls must be retained. For $48 \le g \le 57$, C_1 wins; for $59 \le g \le 80$, C_2 wins alone. The smallest integer guarantee is therefore $g = 59$ hundred USD, or \$5,900. The increase from 48 is $(59 - 48) \times \$100 = \$1,100$ in high-attendance profit.

At $g = 59$, C_2 has payoffs $(59, 32)$, row maximum 59, row minimum 32, maximum regret 21, and

$$EMV(C_2) = 0.4(59) + 0.6(32) = 23.6 + 19.2 = 42.8.$$

C_1 remains unchanged: maximum 80, minimum 10, maximum regret 22, and EMV $0.4(80) + 0.6(10) = 38$. All four updated recommendations are:

Criterion	Recommended alternative(s)	Deciding statistic and value
Maximax	C_1	Row maximum: 80
Maximin	C_2	Row minimum: 32
Minimax Regret	C_2	Maximum regret: 21
EMV/Bayes	C_2	EMV: 42.8

There are no ties at $g = 59$; the boundary tie at $g = 58$ belongs to Minimax Regret only.



3 Book Fair Stock [C9-2]

A bookseller chooses a fair stock plan D_1–D_6. Net profits are in hundred USD under high, normal, and low sales, with probabilities 0.5, 0.3, and 0.2.

Alternative	Max	Min	Regret	EMV
D_1	100	0	40	65
D_2	90	10	30	65
D_3	80	20	20	65
D_4	70	30	30	59
D_5	60	40	40	53
D_6	50	30	50	43

Use the supplied summary statistics directly, including the original maximum regrets; no statistic is recalculated after filtering. Compare only survivors at each stage and retain all cutoff ties.

1. EMV/Bayes: keep 2 highest EMVs.
2. Maximin: select highest Min.

Questions/Tasks

C9 Construct the filtering tree and give a one-sentence interpretation. **[1]**

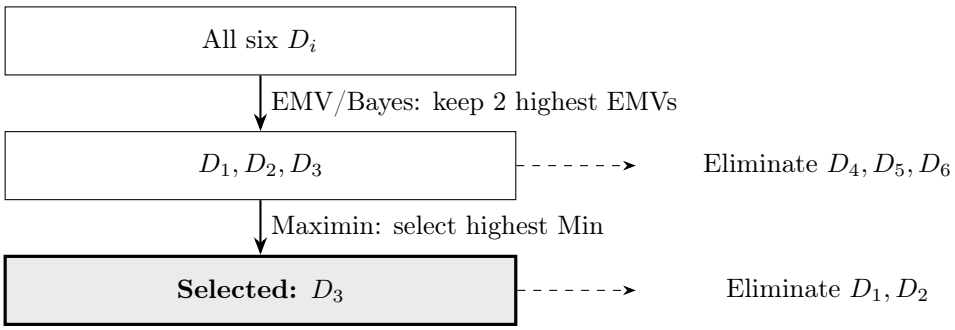
C9 Construct a supporting table with rules, values, survivors, eliminations, and final choice. **[1]**

Supporting filtering table [1]

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$D_1, D_2, D_3, D_4, D_5, D_6$	—
1	EMV/Bayes: keep 2 highest EMVs	$D_1(65)$, $D_2(65)$, $D_3(65)$	$D_4(59)$, $D_5(53)$, $D_6(43)$
2	Maximin: select highest Min	**Selected:** $D_3(20)$	$D_1(0)$, $D_2(10)$

The second-ranked EMV is 65, shared by D_1, D_2, D_3, so all three survive the first stage. The final comparison uses only their minima $0, 10, 20$.

Filtering tree and interpretation **[1]**



The book fair selects D_3: it shares the highest EMV of 65 and has the highest minimum profit, 20 hundred USD, among the retained plans.

4 Refill Shop Expansion [C9-2]

A refill shop chooses next year's expansion plan F_1–F_6. Net profits are in thousand USD under strong, steady, and weak demand, with probabilities 0.3, 0.5, and 0.2.

Alternative	Max	Min	Regret	EMV
F_1	120	0	45	61
F_2	105	15	30	64.5
F_3	95	25	25	68.5
F_4	85	35	35	70
F_5	75	45	45	64
F_6	65	40	55	55

Use the supplied summary statistics directly, including the original maximum regrets; no statistic is recalculated after filtering. Compare only survivors at each stage and retain all cutoff ties.

1. Minimax Regret: keep 5 lowest.
2. EMV/Bayes: keep 4 highest.
3. Maximax: keep 2 highest Max.
4. Maximin: select highest Min.

Questions/Tasks



C9 Construct the filtering tree and give a one-sentence interpretation. **[1]**

C9 Construct a supporting table with rules, values, survivors, eliminations, and final choice. **[1]**

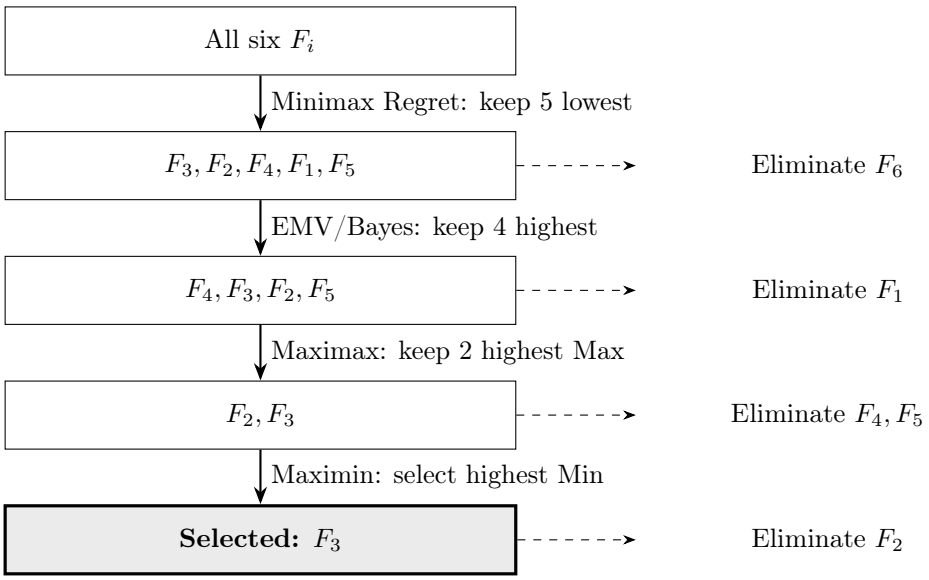
Supporting filtering table [1]

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$F_1, F_2, F_3, F_4, F_5, F_6$	—
1	Minimax Regret: keep 5 lowest	$F_3(25)$, $F_2(30)$, $F_4(35)$, $F_1(45)$, $F_5(45)$	$F_6(55)$
2	EMV/Bayes: keep 4 highest	$F_4(70)$, $F_3(68.5)$, $F_2(64.5)$, $F_5(64)$	$F_1(61)$
3	Maximax: keep 2 highest Max	$F_2(105)$, $F_3(95)$	$F_4(85)$, $F_5(75)$
4	Maximin: select highest Min	**Selected:** $F_3(25)$	$F_2(15)$

At the first cutoff, F_1 and F_5 both have regret 45, and both survive. Later cutoffs have no ties.



Filtering tree and interpretation **[1]**



The refill shop selects F_3: after the regret, EMV, and upside filters, its minimum profit of 25 thousand USD exceeds F_2's 15.

5 School Supply Orders [C9-2]

A school supplier chooses next term's order plan H_1–H_6. Net profits are in hundred USD under high, normal, and low demand, with probabilities 0.3, 0.5, and 0.2.

Alternative	Max	Min	Regret	EMV
H_1	120	0	45	61
H_2	105	15	30	64.5
H_3	95	25	25	68.5
H_4	85	35	35	70
H_5	75	45	45	64
H_6	65	40	55	55

Use the supplied summary statistics directly, including the original maximum regrets; no statistic is recalculated after filtering. Compare only survivors at each stage and retain all cutoff ties.

1. Maximin: keep 4 highest Min.
2. Minimax Regret: keep 3 lowest.
3. Maximax: keep 2 highest Max.
4. EMV/Bayes: select highest EMV.

Questions/Tasks

C9 Construct the filtering tree and give a one-sentence interpretation. **[1]**

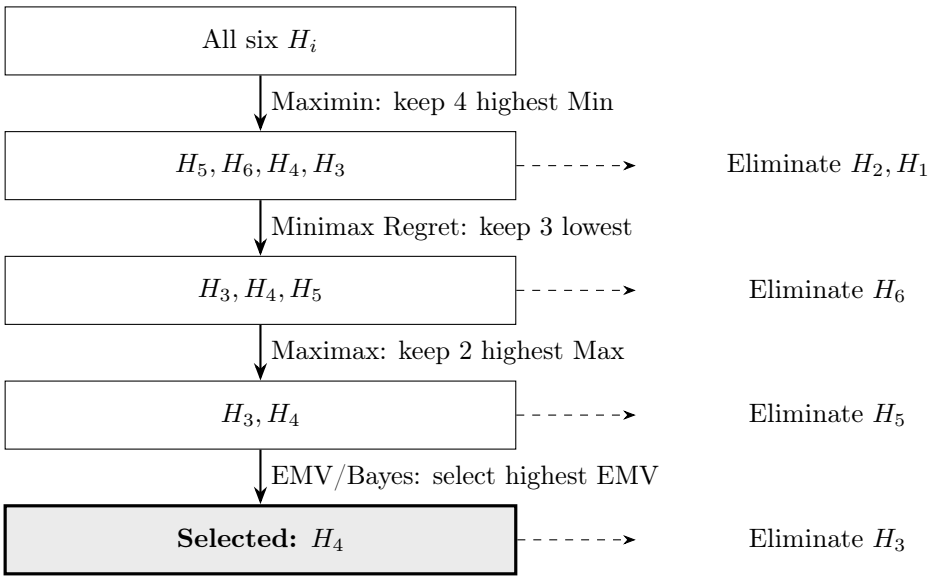
C9 Construct a supporting table with rules, values, survivors, eliminations, and final choice. **[1]**

Supporting filtering table [1]

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$H_1, H_2, H_3, H_4, H_5, H_6$	—
1	Maximin: keep 4 highest Min	$H_5(45)$, $H_6(40)$, $H_4(35)$, $H_3(25)$	$H_2(15)$, $H_1(0)$
2	Minimax Regret: keep 3 lowest	$H_3(25)$, $H_4(35)$, $H_5(45)$	$H_6(55)$
3	Maximax: keep 2 highest Max	$H_3(95)$, $H_4(85)$	$H_5(75)$
4	EMV/Bayes: select highest EMV	**Selected:** $H_4(70)$	$H_3(68.5)$

The cutoffs are Min 25, maximum regret 45, and Max 85; the final EMV comparison is 70 versus 68.5. There are no cutoff or final ties.

Filtering tree and interpretation **[1]**



The school supplier selects H_4: after the downside, regret, and upside filters, its EMV of 70 hundred USD exceeds H_3's 68.5.

C10

C10 example solution.